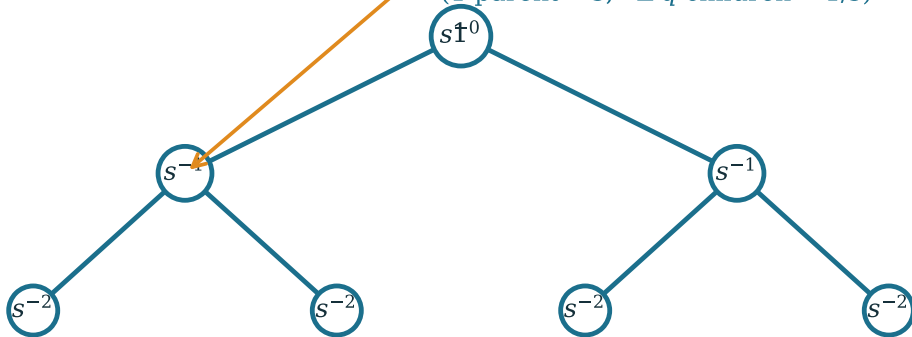


Why the ceiling is exactly $2\sqrt{\Delta - 1}$ (weighted Gershgorin on the tree)

$$\text{row sum at } v: 1 \cdot s + q \cdot \frac{1}{s} = s + \frac{q}{s} = 2\sqrt{q}$$

(1 parent $\times s$, $\leq q$ children $\times 1/s$)



weight $w_v = (\sqrt{\Delta - 1})^{-\text{dist}(r, v)}$, $s = \sqrt{\Delta - 1} = \sqrt{q}$: branching q balanced by decay $\sqrt{q}$ per level